

Research Paper

Antifungal effect of hydroethanolic extract of *Fridericia chica* (Bonpl.) L. G. Lohmann leaves and its therapeutic use in a vulvovaginal candidosis model



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ABSTRACT

Vulvovaginal candidosis (VVC) is one of the most frequent causes of gynecological consultations. Therefore, the development of new antifungal therapies against VVC is relevant. In this context, the leaves of *Fridericia chica* (Bonpl.) L. G. Lohmann are considered a therapeutic alternative since they are traditionally used in VVC therapy. However, no scientific evidence has supported this use against fungal vaginal infections. Then, we aimed to characterize the antifungal effect of a hydroethanolic extract of *F. chica* leaves (HEFc) and evaluate the therapeutic potential of this extract in a VVC model. HEFc inhibited the growth of *C. albicans* (256–1,204 $\mu\text{g/mL}$) and *C. krusei* (512 $\mu\text{g/mL}$) in vitro. HEFc inhibited yeast-to-hypha transition in *C. albicans* and has a low ability to induce resistance in this species. Intravaginal use of the HEFc at 50 mg/mL causes mycological cure in animals with VVC after 6 days of treatment, similar to the effect observed for the commercial antifungal nystatin. These results support the traditional use of *F. chica* leaves as a topical therapeutic option to treat VVC.

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Introduction

Vulvovaginal candidosis (VVC) is an acute inflammatory disease caused by the uncontrolled proliferation of *Candida* species (especially *C. albicans*) in the vaginal tissue [1]. The fungal vaginitis by *Candida* is the second most common vaginal infection in North America and the most prevalent human *Candida* infection. For instance, it is estimated that 75% of women will have at least one episode of candidiasis in their lifetime, and 40%–45% will have two or more episodes [2]. Moreover, approximately 10%–20% of women will have complicated cases of VVC, which includes severe VVC, persistent non-albicans *Candida* (NACs) infection, and recurrent VVC [1].

Abbreviations: VVC, Vulvovaginal candidosis; NAC, non-albicans *Candida*; HEFc, Hydroethanolic extract of *Fridericia chica* leaves; MIC, Minimum inhibitory concentration; MBC, Minimum bactericidal concentration; FBS, Fetal bovine serum; SDB, Sabouraud-Dextrose broth; SDA, Sabouraud-Dextrose agar; CFU, Colony-forming unit; ANOVA, Analysis of variance; Y-H, Yeast-to-hypha

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Importantly, VVC has received limited attention from public health authorities, funding agencies, researchers, and pharmaceutical companies worldwide. As a result, the therapeutic options for the treatment of VVC are limited, and little has been done to expand this antifungal arsenal [3]. Furthermore, although *Candida albicans* has been historically known as the main causative agent of VVC, new evidence reveals an increasing incidence of NACs, such as *C. glabrata*, *C. krusei*, and *C. tropicalis*. These species are known to rapidly develop resistance to antifungal agents, which may contribute to complications of VVC [4]. In this context, the development of new antifungal strategies against VVC becomes imperative [5].

Natural products are currently the main source of agents with antifungal properties [6]. Plants possess several classes of bioactive compounds, such as terpenoids, alkaloids, flavonoids, and steroids, with known fungistatic or fungicidal effects [7]. In this regard, the leaves of *Fridericia chica* (Humb. & Bonpl.) B. Verlot (Bignoniaceae) (previously known as *Arrabidaea chica*), a plant native to the tropical regions of Africa and South America, have been traditionally used in the treatment of uterine inflammation and genitourinary infections,

including VVC [8]. Popularly known in Brazil as “crajiuru,” “cipó-cruz,” or “guajuru-piranga” (in Portuguese), *F. chica* is a woody, shrubby, and climbing vine that has been used for body painting rituals and as sunscreen and insect repellents by native communities [8]. Several studies have confirmed the traditional uses of this species, including its antimicrobial [9] and anti-inflammatory activities [10–12].

The pharmacological properties of *F. chica* are associated with the presence of phenolic compounds in the leaves, mainly flavonoids (kaempferol, thevetiaflavone, carajuflavone, vicentin-2, acacetin, 4',6,7-trihydroxy-5-methoxyflavone, and luteolin) and anthocyanidins (3-deoxyanthocyanidins) [13,14]. Many of these phenolic compounds possess antifungal and anti-inflammatory effects, which can be beneficial in the therapeutic management of VVC [12,13,15]. However, there is currently no scientific evidence of the efficacy of the extract obtained from *F. chica* leaves against VVC. Thus, we aimed to characterize the antifungal effect of a hydroethanolic extract of *F. chica* leaves (HEFc) in vitro and in a VVC model using rats.

Methods

Microorganisms

The microorganisms used in this study were originated from the American Type Culture Collection (ATCC). The antimicrobial activity was investigated against *C. albicans* ATCC 10231, *C. albicans* ATCC 18804, *C. albicans* ATCC 2091, *C. albicans* ATCC MYA-2876 (SC 5314), *C. glabrata* ATCC 2001, *C. krusei* ATCC 34135, and *C. tropicalis* ATCC 28707. The antifungal activity was also determined against seven clinical isolates of *Candida albicans* originated from vaginal secretions of women with VVC. All clinical isolates were kindly provided by Jaqueline M. S. Ferreira (Laboratory of Medical Microbiology, Universidade Federal de São João Del Rei, Divinópolis, MG, Brazil) and were identified by matrix-assisted laser desorption ionization-time of flight mass spectrometry (MALDI-TOF MS), using a Microflex LT spectrometer (BrukerDaltonics, MA, USA).

Reagents and extracts

Amphotericin B (Inlab, São Paulo, SP, Brazil), miconazole (Pharma Nostra, Rio de Janeiro, RJ, Brazil), nystatin (Fragon, São Paulo, SP, Brazil), crystal violet, 95% ethanol, sodium chloride (NaCl) (Synth, São Paulo, SP, Brazil), glucose (Inlab, São Paulo, SP, Brazil), ergosterol (Sigma-Aldrich, San Francisco, CA, USA), and fetal bovine serum (FBS) (Cultilab, São Paulo, SP, Brazil) were purchased from commercial suppliers and used without additional purification. Sabouraud-Dextrose broth (SDB) and Sabouraud-Dextrose agar (SDA) were purchased from Kasvi (São José do Pinhais, PR, Brazil). The hydroethanolic extract (ethanol:water; 70:30) obtained from *F. chica* leaves (HEFc) was provided by the Pharmacognosy Laboratory, Department of Pharmaceutical Products, Federal University of Minas Gerais, Belo Horizonte, MG, Brazil. This extract was obtained and chemically standardized according to the procedure described by Takenaka et al [16]. HEFc was reconstituted in sterile deionized water and maintained at -20°C until use.

Antifungal activity

The antifungal activity was determined by the broth microdilution method as described by Andrade et al [17]. Twenty microliters (20 μL) of HEFc was two-fold serially diluted (2–1,024 $\mu\text{g/mL}$) in microplates containing 180 μL of SDB (Acumedia, Brazil). Subsequently, 100 μL of a *Candida* inoculum at 10^3 colony-forming unit (CFU)/mL prepared in saline solution from a 48 h culture was added to each well. The plates were incubated at $35\pm 2^{\circ}\text{C}$ for 48 h. Then, the minimum inhibitory concentration (MIC) was defined as the lowest concentration of extract that inhibited the visible growth of the yeast.

The minimum fungicidal concentration (MFC), in turn, was determined by the plating of 10 μL of samples from each well that did not show any visible growth in the MIC assay [18]. After incubation at $35\pm 2^{\circ}\text{C}$ for 48 h, the MFC was defined as the lowest concentration of extract that inhibits the growth of colonies on the SDA.

Time-kill curve

The effect of the extract as a function of exposure time was studied as described by Andrade et al [17]. The yeast was challenged with HEFc at 1x (512 $\mu\text{g/mL}$), 1/2x (256 $\mu\text{g/mL}$), and 1/4x (128 $\mu\text{g/mL}$) MIC, and the fungal burden (CFU/mL) was determined at 0, 1, 3, 6, 9, 12, 24, and 48 h.

Phenotypic effects on fungal cells

Sorbitol assay: The antifungal activity of the extract against *C. albicans* was determined by the broth microdilution procedure using Sabouraud-Dextrose broth supplemented with 0.8 M of sorbitol [18]. The MIC values were determined after 2 and 7 days of incubation at $35\pm 2^{\circ}\text{C}$ and the HEFc was tested at concentration range of 4–2,048 $\mu\text{g/mL}$. Caspofungin was used as the positive control.

Exogenous ergosterol binding assay: Antifungal activity of HEFc against *C. albicans* was determined in the presence or absence of increasing concentrations of exogenous ergosterol (50–250 $\mu\text{g/mL}$) [18]. The MIC values were determined after 48 h of incubation at $35\pm 2^{\circ}\text{C}$ and the HEFc was tested at concentration range of 4–2,048 $\mu\text{g/mL}$. Nystatin was used as the positive control.

Yeast-to-hypha transition: The effect of HEFc on the yeast-to-hypha transition of *C. albicans* was also evaluated. Hyphal induction was conducted by incubating *C. albicans* ATCC 10231 (10^3 CFU/mL) in microplates containing SDB supplemented with FBS at 50% v/v [17]. Posteriorly, HEFc at 512 or 1,024 $\mu\text{g/mL}$ was added to the growth medium. The microplates were incubated for 24, 48, and 72 h at $35\pm 2^{\circ}\text{C}$, and the hypha formation was observed by a light microscope (Nikon TE 2000-U Eclipse) with a magnification of 400x.

Multi-step resistance study

The ability of *C. albicans* ATCC 10231 to acquire resistance to HEFc was investigated through a multi-step resistance study, as described by Bogdanovich et al [19], with modifications. Glass tubes containing 1 mL of HEFc-free or HEFc-supplemented SDB (HEFc at 128 $\mu\text{g/mL}$: $1/4 \times \text{MIC}$) were inoculated with 10 μL of a fungal suspension at 10^6 CFU/mL. The tubes were incubated at $35\pm 2^{\circ}\text{C}$ for 24 h prior to each serial passage. Passages were performed at 24 h intervals for up to 21 days by transferring 10 μL of the 24 h culture to a tube containing fresh culture medium. Daily, a streak of each culture was made in SDA, and the MIC was determined according to the method described above. The results were graphically expressed as a function of the time to increase the MIC vs. exposure time (in days). Miconazole was used as the positive control.

In vivo assay in a VVC model

Eight-week-old female Wistar rats (100–150 g) (Biotério Central da UFSJ, São João Del Rei, MG, Brazil) were used in this study. The VVC model was established according to the protocol described by Yano and Fidel [20]. First, the pseudo-estrous state was induced by subcutaneous administration of 0.2 mg/mL of estradiol cypionate of 24/24 h during 5 days before infection and 48/48 h after infection. Then, the rats were immunosuppressed by administering one dose of 50 mg/kg of cyclophosphamide intraperitoneally (i.p.). On the first day, 0.1 mL of viable *C. albicans* ATCC 10231 suspension (5.0×10^7 CFU/mL) was inoculated into the vaginal cavity of the rats. Two days after infection (day 0), the vaginal fungal load was evaluated through

lavage with 0.1 mL of sterile saline and determined in SDA culture supplemented with 50 mg/L chloramphenicol, in which the results were expressed as CFU/mL. The infection was considered satisfactory if the count for the vaginal lavage cultures of each rat was 10^2 CFU/mL.

After confirmation of the vaginal infection, the rats were randomized equally in six groups ($n = 6$ in each group): non-infected (1); infected and untreated (2); infected and treated with vaginal nystatin cream (Medley, Brazil) (3); and infected and treated with HEFc at 0.5 mg/mL (4), 5 mg/mL (5), and 50 mg/mL (6). Treatments, vehicle, and positive control were administered intravaginally to the infected animals once a day for 6 consecutive days. On days 0, 3, and 6 after the establishment of the infection, the vaginal fungal load was evaluated to verify the effectiveness of the treatments. All experimental procedures were approved by the Animal Experimentation Ethics Committee of the Universidade Federal de Minas Gerais (Protocol UFMG/CEUA 121/2021).

Statistical analyses

All in vitro tests (except time-kill curve) were performed in triplicate with at least three independent experiments. The normality of data was assessed using the Shapiro-Wilk test and descriptive statistics analysis. Phenotypic assays were analyzed using one-way analysis of variance (ANOVA) followed by the Dunnett post-hoc test, with the data being compared to the untreated control. In the VVC model, differences of fungal load between groups were assessed using the ANOVA test followed by Tukey post-hoc test. A p -value < 0.05 was considered statistically significant. Descriptive and analytical statistics were performed using GraphPad Prism 5.0® (GraphPad Software, Inc., San Diego, CA).

Results and discussion

HEFc exhibits moderate antifungal activity against *Candida*

The antifungal activity of HEFc was evaluated through the determination of the MIC. As shown in Table 1, the extract inhibited the growth of all *C. albicans* strains tested with MIC ranging from 256 to 1,204 μ g/mL. Among the NACs studied, only *C. krusei* (512 μ g/mL) was susceptible to the antifungal effect of HEFc. On the other hand, *C. glabrata* and *C. tropicalis* were intrinsically resistant (MIC $> 1,024$ μ g/

mL) to this extract. Although the pathogenicity of non-*albicans Candida* is lower than that of *C. albicans*, infections by these species (mainly *C. glabrata*) often evolve to VVC refractory to conventional therapies, constituting an important clinical challenge [21]. However, our data reveal that HEFc would not have a relevant effect on treating VVC infections complicated by persistent non-*Candida albicans*.

According to Holetz et al [22], the antimicrobial activity of plant extracts is considered good if the MIC values are lower than 128 μ g/mL, moderate if the MIC range is 128 to 512 μ g/mL, weak if the inhibitory concentration is between 513 and 1,024 μ g/mL, and inactive if it is higher than 1,024 μ g/mL. Therefore, HEFc showed a moderate antifungal effect against *C. albicans* and *C. krusei*. It is noteworthy that the extract was active against *C. krusei*, which is intrinsically resistant to azoles, and *C. albicans* ATCC 10231, which has acquired resistance to azoles and echinocandins [17]. In addition to its fungistatic activity, HEFc showed a predominately fungicidal (MFC) effect against *C. albicans* (512 to 1,024 μ g/mL). However, it did not eliminate the cells of *C. krusei* in vitro (Table 1).

Other studies have also indicated the antifungal effect of extracts and fractions obtained from *F. chica* leaves. In agreement with our results, Höfling et al [15], demonstrated that the dichloromethane extract of *F. chica* leaves has a potent antifungal effect against several *Candida* species, including *C. albicans* (0.015 mg/mL), *C. dubliniensis* (0.03 mg/mL), *C. parapsilosis* (0.015 mg/mL), *C. tropicalis* (0.015 mg/mL), *C. guilliermondii* (0.015 mg/mL), *C. krusei* (0.007 mg/mL), *C. lusitanae* (0.007 mg/mL), and *C. rugosa* (0.007 mg/mL), being inactive only against *C. utilis* and *C. glabrata*. In addition, Ribeiro et al [9], showed that the crude ethanolic extract obtained from the *F. chica* leaves at high concentration (500 mg/mL) inhibited the growth of *C. albicans* in the Kirby-Bauer assay (Inhibition halo: 10 mm). However, no inhibition was observed when the extract was tested at concentrations below 250 mg/mL. In contrast, Mota [23] demonstrated that different extracts (aqueous, chloroform, and ethanolic) and fractions (ethyl acetate and butanolic) obtained from *F. chica* leaves (all tested at 50 mg/mL) were active against the dermatophyte *Microsporum canis* (Inhibition halo: 10 mm: 16.1–20.1 mm), but did not inhibit the growth of yeast-like fungi, such as *Candida albicans* and *Malassezia pachydermatis*. Furthermore, the hydroethanolic extract of *F. chica* leaves presented MIC > 800 μ g/mL against *C. albicans*, *C. krusei*, *C. tropicalis*, and *C. parapsilosis* in the study conducted by Mafioleti et al [24], revealing the absence of anti-*Candida* effect. This discrepancy in the results may reflect the differences in the experimental conditions,

Table 1

Minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC) of hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leaves against *Candida* spp.

Microorganisms	Extract/Compounds (μ g/mL)							
	HEFc		Miconazole		Nystatin		Amphotericin B	
	MIC	MFC	MIC	MFC	MIC	MFC	MIC	MFC
<i>Candida albicans</i> ATCC 10231*	512	1,024	8	8	4	4	1	1
<i>Candida albicans</i> ATCC 18804	512	1,024	8	8	4	4	0.5	1
<i>Candida albicans</i> ATCC 2091	512	512	4	4	2	2	0.5	0.5
<i>Candida albicans</i> SC 5314	512	512	4	4	2	4	1	1
<i>Candida glabrata</i> ATCC 2001	$> 1,024$	–	0.25	0.25	2	4	0.5	0.5
<i>Candida tropicalis</i> ATCC 28707	$> 1,024$	–	0.12	0.25	2	2	0.25	1
<i>Candida krusei</i> ATCC 34135	512	$> 1,024$	4	4	16	32	0.5	1
<i>Candida albicans</i> CI 1 [#]	512	$> 1,024$	8	8	2	2	0.5	1
<i>Candida albicans</i> CI 2 [#]	512	1,024	4	16	4	4	0.5	1
<i>Candida albicans</i> CI 3 [#]	512	512	8	8	4	4	1	1
<i>Candida albicans</i> CI 4 [#]	1,024	$> 1,024$	8	32	4	4	1	1
<i>Candida albicans</i> CI 5 [#]	256	512	2	2	2	2	1	1
<i>Candida albicans</i> CI 6 [#]	512	$> 1,024$	4	4	2	4	1	2
<i>Candida albicans</i> CI 7 [#]	512	$> 1,024$	2	2	8	8	1	1

HEFc: Hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leave.

* *Candida albicans* ATCC 10231 is resistant to Anidulafungin, Voriconazole, Itraconazole, and Fluconazole.

[#] All clinical isolates were recovered from vaginal discharge from women with a positive diagnosis of vulvovaginal candidiasis.

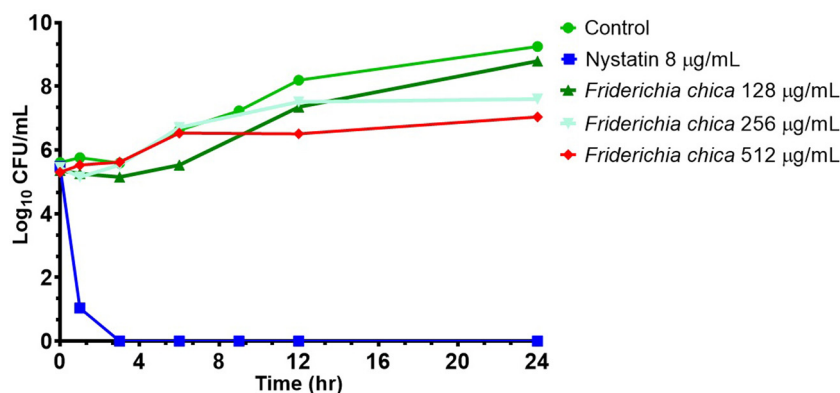


Fig. 1. Time-Kill curve assay of the hydroethanolic extract of *F. chica* leaves (HEFc) against *Candida albicans* ATCC 10231. HEFc was evaluated at 128 µg/mL, 256 µg/mL, and 512 µg/mL. Nystatin (8 µg/mL) and untreated fungal cells were employed as the positive and growth controls, respectively.

including the origin of the plant material employed, types of extracts and extraction methods performed, *Candida* lines and species used, and method of evaluation of the antifungal effect selected. Indeed, most of the previous studies do not use the broth microdilution method to assess antifungal activity.

HEFc reduces the growth rate of an azole-resistant *Candida albicans*

An alternative approach to assess the efficacy of antifungals is to use dynamic in vitro models based on time-kill curves. Time-kill curves follow microbial killing or growth as a function of time and antifungal concentration, allowing a detailed assessment of the pharmacokinetic-pharmacodynamic relationship [25]. Here, we performed the time-kill curve assay to evaluate the effect of the exposure time and HEFc concentration on the growth kinetics of *C. albicans*. As shown in Fig. 1, cells treated with HEFc at 256 µg/mL and 512 µg/mL presented a stable growth after 12 h of incubation, while the untreated suspensions continued to increase their cell density throughout the experimental period evaluated. For instance, after 24 h of incubation, the fungal load in the HEFc treated tubes at 256 µg/mL (7.6 Log₁₀CFU/mL) and 512 µg/mL (7.04 Log₁₀ CFU/mL) is considerably lower than in the control tube (9.25 Log₁₀ CFU/mL). Thus, the HEFc-induced delay on fungal growth can assist in the VVC therapy by ensuring that the immune system of the vaginal tissue has time to fight the infection [26]. Nystatin (8 µg/mL), in turn, killed all fungal cells after 2 h of incubation, confirming its potent fungicidal effect and validating our experimental conditions.

HEFc inhibits the yeast-to-hypha transition

C. albicans is characterized by its morphological plasticity, in which it adopts two major forms: the ovoid yeast and elongated hypha [17,18]. Yeast-to-hypha (Y-H) transition plays essential roles in the pathophysiology of candidiasis and is considered one of the most important virulence factors of *C. albicans*. In this context, the effect of HEFc on the Y-H transition was evaluated. As shown in Fig. 3, the extract appears to have reduced the number of fungal hyphal structures after exposure to FBS compared to the untreated cells. This effect was more pronounced at 1.024 µg/mL but was reduced in higher incubation periods (especially after 72 h of incubation) (Fig. 3). These results suggest that HEFc decelerates the fungal filamentation, impairing the ability of *C. albicans* to adapt to the host and, consequently, its capacity to infect the vaginal tissue.

It is worth noting that in the antifungal activity experiment, HEFc at 1,024 µg/mL was fungicidal, but the growth of fungal cells was observed in the yeast-to-hypha transition assay at this concentration. It could be explained by the addition of fetal bovine serum to the

culture medium in the latter case, which has been reported to reduce the antifungal effect of some agents in previous studies [27,28]. In fact, in the yeast-to-hypha transition assay, it was observed that even in the wells treated with HEFc at 1,024 µg/mL, there was visible fungal growth comparable to the growth of untreated wells. Therefore, in the presence of fetal bovine serum, the antifungal activity of the extract was considerably compromised, suggesting that the effect of HEFc on the filamentation of *C. albicans* is not associated with the reduction of fungal viability but with the inhibition of mechanisms involved in hypha formation (e.g., initiation of germ tube, cell elongation, hyphal extension/growth).

Phenotypic effects of HEFc on *Candida albicans*

Despite the previous evidence regarding the antifungal properties of *F. chica* leaves, so far, there is no data on the potential mechanism of action of this plant species. Furthermore, the effect of *F. chica* on phenotypic characteristics relevant to the pathogenesis of fungi remains to be elucidated. Initially, we determined the effect of HEFc on the cell membrane by studying the ability of compounds present in the extract to bind to ergosterol. In this direction, the MIC of HEFc against *C. albicans* remained unchanged after the addition of increased concentrations of exogenous ergosterol in the culture medium (Fig. 2), indicating that the antifungal activity of HEFc is not effected by direct ergosterol binding. In contrast, the antifungal activity of nystatin and amphotericin B, two antifungals that act on the cell membrane through ergosterol binding, decreased in the presence of exogenous ergosterol, validating our experimental conditions (Fig. 2).

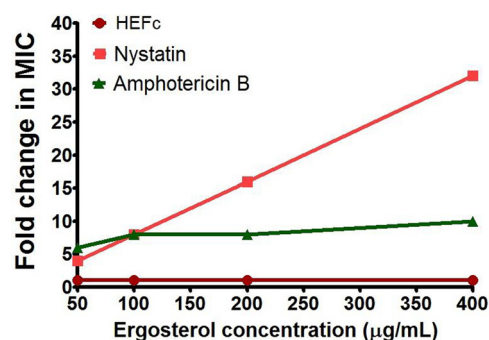


Fig. 2. Effect of exogenous ergosterol addition (50–400 µg.mL⁻¹) on the minimum inhibitory concentration (MIC) of the hydroethanolic extract of *F. chica* leaves (HEFc) against *Candida albicans* ATCC 10231.

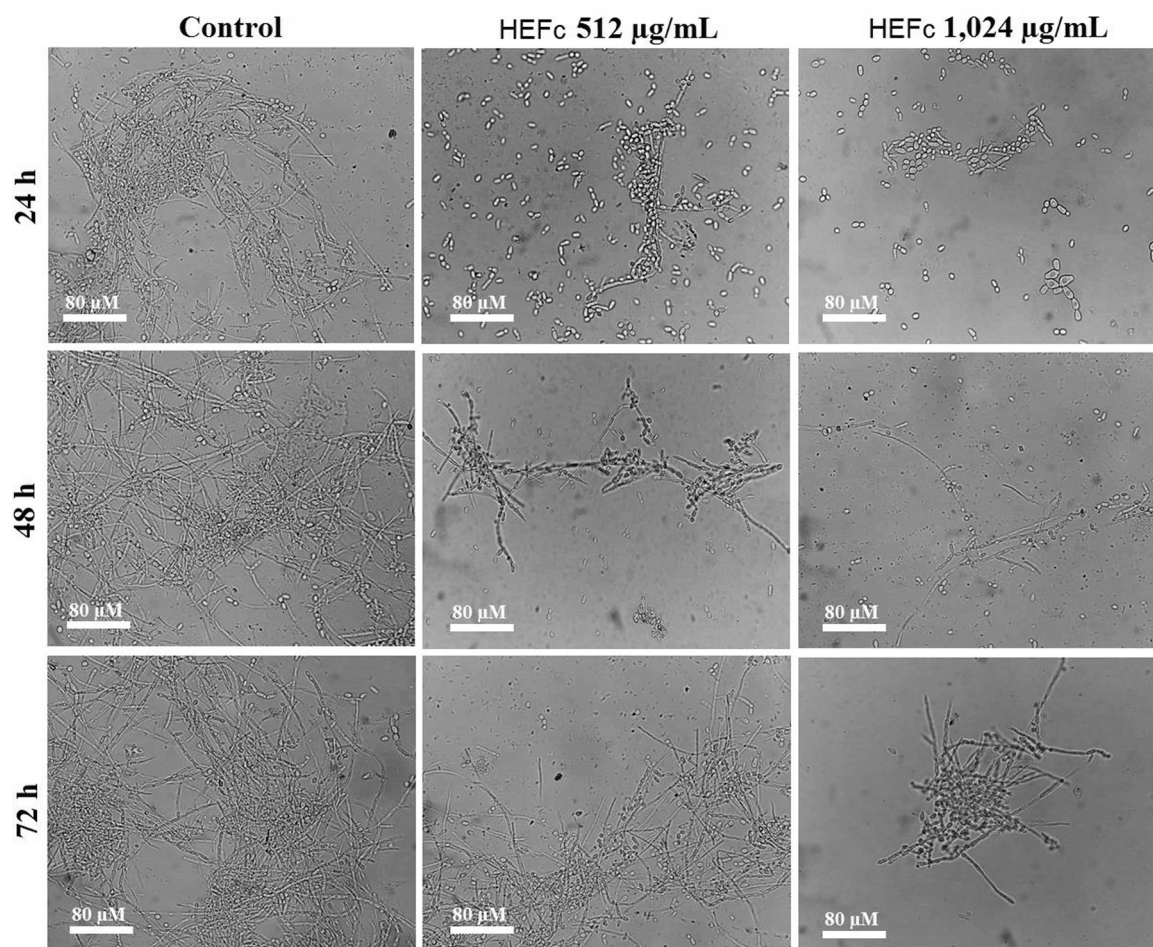


Fig. 3. Yeast-to-hypha transition in *Candida albicans* ATCC 10,231 after treatment with hydroethanolic extract of *F. chica* leaves (HEFc). *C. albicans* were cultured with HECf for 24, 48, and 72 h at 35 ± 2 °C. Representative microphotographs are shown. The white bar represents a length of 80 µm (magnification of $\times 400$).

Then, the effect of HECf on the fungal cell wall was evaluated. For this, the MIC assay was conducted in the presence of sorbitol, a compound that acts as an osmotic protector of the fungal cell wall. After adding sorbitol to the medium, no change in the MIC values of HECf against *C. albicans* was observed, indicating no effect on the fungal cell wall (Table 2). Caspofungin, a drug that induces structural alterations in the fungal cell wall, showed increased MIC against *C. albicans* after sorbitol supplementation.

HEFc does not induce fungal resistance in vitro

Currently, fungal resistance is the biggest challenge for the pharmaceutical industry. The rate of new antifungal agents development is significantly lower than the ability of microorganisms to become resistant to these therapeutic agents [29,30]. Thus, we assessed

whether HECf induces resistance in *C. albicans* by a multi-step resistance selection study. Repeated exposure to HECf did not generate detectable resistance during the 21 days of passage, indicating that this extract has a low capacity of fungal resistance induction (Fig. 4). Indeed, the chemical complexity of plant extracts is an important barrier to microbial resistance. The chemical components of plants exert their antifungal activity through multiple mechanisms, which hinders the microorganism's ability to become resistant against all these agents [6].

Topical use of the HECf induces mycological cure in a VVC model

One of the most traditional uses of *F. chica* leaves is in the therapeutic management of VVC, which has not been confirmed by scientific evidence yet [8]. Based on the in vitro results obtained for the antifungal effect of HECf, we evaluated its therapeutic effect in an in vivo model of VVC. Previous reports showed that immunosuppressive rats in a prolonged pseudo-estrus state are particularly susceptible to fungal colonization of the vaginal tissue, representing, thus, a good model for the study of the pathogenesis and therapeutics of VVC [20,31]. As shown in Table 3, the animals treated with HECf at 0.5 mg/mL (1.12 ± 0.94 Log₁₀ CFU/mL) and 5 mg/mL (1.08 ± 0.36 Log₁₀ CFU/mL) intravaginally showed a significant reduction in the fungal load ($p < 0.05$) after 6 days compared to the untreated group (3.90 ± 0.36 Log₁₀ CFU/mL) (Table 3). In addition, in the animals treated with the highest concentration of the extract (50 mg/mL), it was possible to verify a complete mycological cure, similarly to what occurred in

Table 2

Minimum inhibitory concentration (MIC; µg/mL) of hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leaves against *Candida albicans* ATCC10231 in the absence or presence of 0.8 M sorbitol.

Extract/Control	2 days		7 days	
	-/Sorbitol	+/-Sorbitol	-/Sorbitol	+/-Sorbitol
HEFc	512	512	512	512
Caspofungin	4	16	4	32

HEFc: Hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leaves.

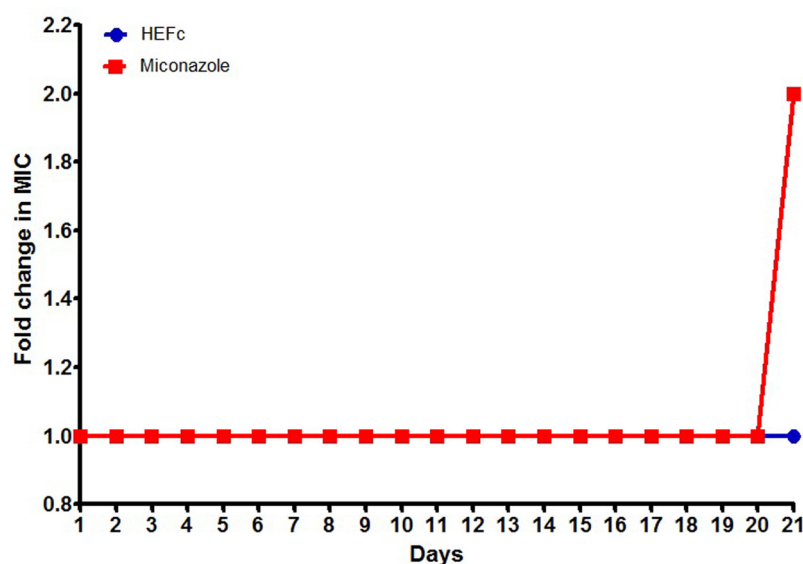


Fig. 4. Multi-step resistance study of the hydroethanolic extract of *F. chica* leaves (HEFc) against *Candida albicans* ATCC 10231. The yeast was serially passaged over 21 days, and the broth microdilution assay was used to determine the minimum inhibitory concentration (MIC) of each compound after each successive passage.

Table 3

Number of infected animals and quantification of the fungal load observed in animals infected with an azole-resistant *C. albicans* (ATCC 10231) strain and treated intravaginally with different concentrations of hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leaves.

Groups	Day 0		Day 3		Day 6		Reduction ^a (%)
	Log ₁₀ CFU/mL ± SD	Infected animals/N	Log ₁₀ CFU/mL ± SD	Infected animals/N	Log ₁₀ CFU/mL ± SD	Infected animals/N	
Nystatin 25,000 UI/g	4.07 ± 0.47	5/5	0.56 ± 0.00	1/5	0.00 ± 0.00 ^b	0/5	100
Infection control	4.32 ± 0.06	5/5	4.24 ± 0.09	5/5	3.90 ± 0.36	5/5	—
Vehicle	4.14 ± 0.10	5/5	3.40 ± 0.77	5/5	3.15 ± 0.78	5/5	—
HEFc 0.5 mg/mL	3.89 ± 0.06	5/5	2.70 ± 0.60	5/5	1.12 ± 0.94 ^b	3/5	71.20
HEFc 5.0 mg/mL	4.04 ± 0.03	5/5	3.43 ± 0.82	5/5	1.08 ± 0.36 ^b	3/5	73.26
HEFc 50 mg/mL	4.18 ± 0.12	5/5	3.41 ± 0.28	5/5	0.00 ± 0.00 ^b	0/5	100

HEFc: Hydroethanolic extract from *Fridericia chica* (Bonpl.) L.G. Lohmann leaves.

^a Percentage reduction in fungal load compared to the first day of treatment. ^b Statistically significant difference compared to day 0 ($p < 0.05$). Nystatin vaginal cream (25,000 UI/g) was used as positive control. SD = standard deviation. N = 5.

the group treated with nystatin (positive control) (Table 3). The vehicle used (saline) did not show any effect on the fungal load of the vaginal canal, validating our experimental conditions. Taken together, these results confirmed for the first time the potential of *F. chica* leaves as a therapeutic option against VVC. It is worth mentioning that, in addition to the antifungal effect presented in this study, there are several pieces of evidence on the anti-inflammatory potential of the leaves of *F. chica* [10–12,16], which can help control important inflammatory symptoms of VVC, such as vaginal discharge and local itching, burning, pain, and redness [1,2].

The therapeutic effect of HEFc was also monitored by the histopathological analysis of the vaginal canal of rats infected with *C. albicans*. As shown in Fig. 5, the untreated animals presented some granulomas in the epithelial region. Additionally, this process contributed to the epithelial vacuolization observed in infected animals. Treatment with the lowest dose of HEFc (5 mg/mL) did not reverse the deleterious effects of the infection of the vaginal tissue. However, a significant therapeutic effect of HEFc was observed at the higher doses evaluated (5 and 50 mg/mL). Epithelial vacuoles were reduced at the dose of 5 mg/mL and disappeared entirely with the treatment with 50 mg/mL of HEFc. In addition, at 50 mg/mL, a greater integrity of the epithelium was observed, in which its thickness was comparable to the untreated control (not shown). In the nystatin-treated animals, except for a reduction in the thickness of the vaginal epithelium, it was possible to observe the preservation of the mucosa structure compared

to infected animals, validating our experimental conditions. Therefore, these results show that, in addition to mycological cure, HEFc also helps preserve the morphological architecture of the mucosa of the vaginal canal in infected rats, which certainly reflects a reduction in clinical signs and symptoms of VVC.

Study limitation

The intravaginal infection of rats was performed with *C. albicans* ATCC 10231, which was originally isolated from patients with bronchomycosis and is not adapted to infection of the vaginal mucosa [32]. Nevertheless, *C. albicans* ATCC 10,231 has been recently used as a model microorganism in several types of experimental infections, including vulvovaginal candidiasis [31,33,34], candidemia [34], intra-abdominal candidiasis [17,35], and fungal keratitis [36]. Since *C. albicans* ATCC 10231 is proven resistant to some antifungal agents (itraconazole, fluconazole, voriconazole, anidulafungin), has virulence compatible with use in pre-clinical models, and its genome is fully sequenced, the American Type Culture Collection has indicated this pathogen as a model for in vitro and in vivo pre-clinical trials of new antifungal agents [32]. However, we recognize that by not being originally adapted to the vaginal mucosa, *C. albicans* ATCC 10231 may produce a less robust fungal vaginosis in rats than clinical isolates from women with vulvovaginal candidosis.

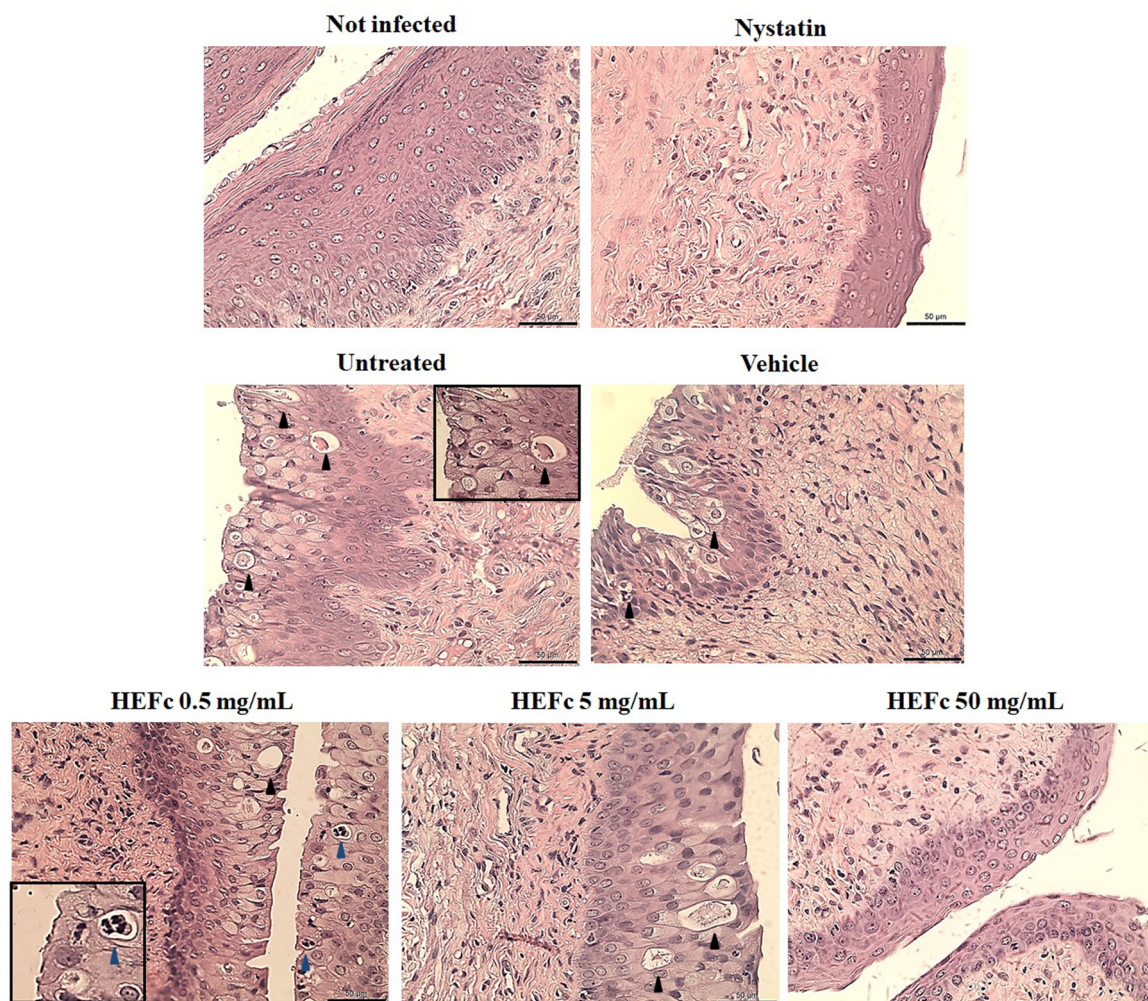


Fig. 5. Histological analysis by hematoxylin and eosin (HE) of the vaginal canal of rats infected with 10^6 CFU/mL of *Candida albicans* ATCC 10231. The animals were treated with HEFc (0.5, 5, and 50 mg/mL), a commercial antifungal (Nystatin 25,000 UI/g), or the vehicle (saline). Black arrow: Epithelial vacuoles; Blue arrow: Granulomas; Original magnifications 400X.

Conclusion

In summary, HEFc presented antifungal effect against the most common pathogen involved in VVC, *C. albicans*, without detectable resistance. In addition, this extract inhibited the yeast-to-hypha transition by *C. albicans*, which are important virulence factors associated with complicated forms of VVC. Finally, we also provided robust evidence about the in vivo effects of HEFc in the treatment of VVC. Therefore, these results corroborate the traditional use of *F. chica* leaves as a topical agent to treat this fungal vaginal infection.

Conflict of Interest

The authors report that they do not have any conflicts of interest.

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